CMAI 480

Computer Vision

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Image Alignment and Stitching

Chapter 8

Reading

- Szeliski: 8.1

Introduction

- Image Alignment is about computing transformations
- Given a set of matches between images A and B
  - How can we compute the transform *H* from A to B?
  - Find transform *H* that best "agrees" with the matches



A



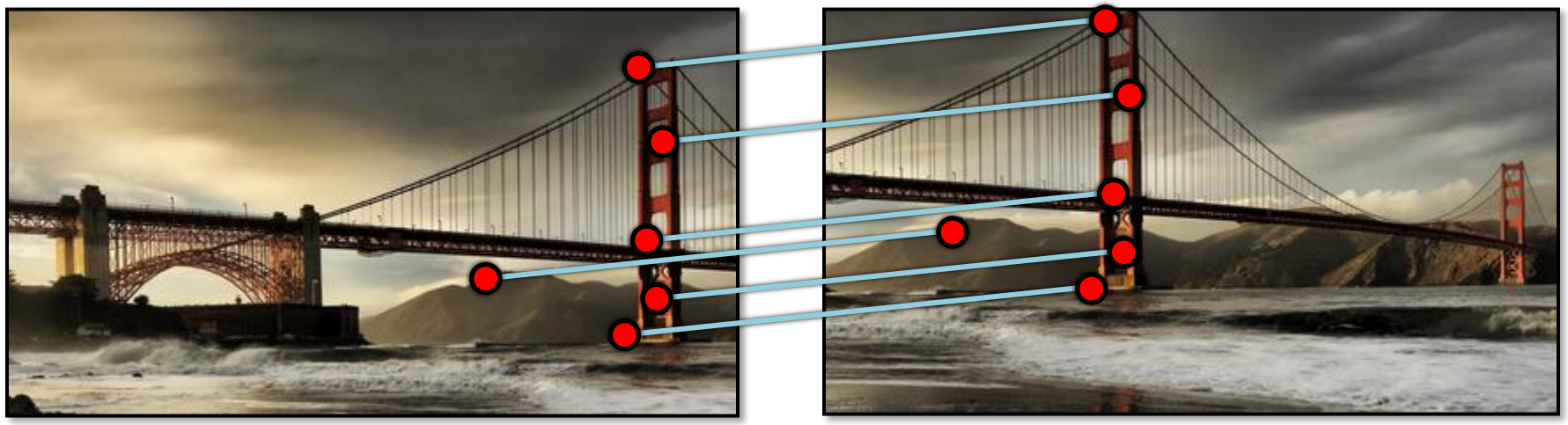
B

?



Pure Translation Case

Simple case: translations

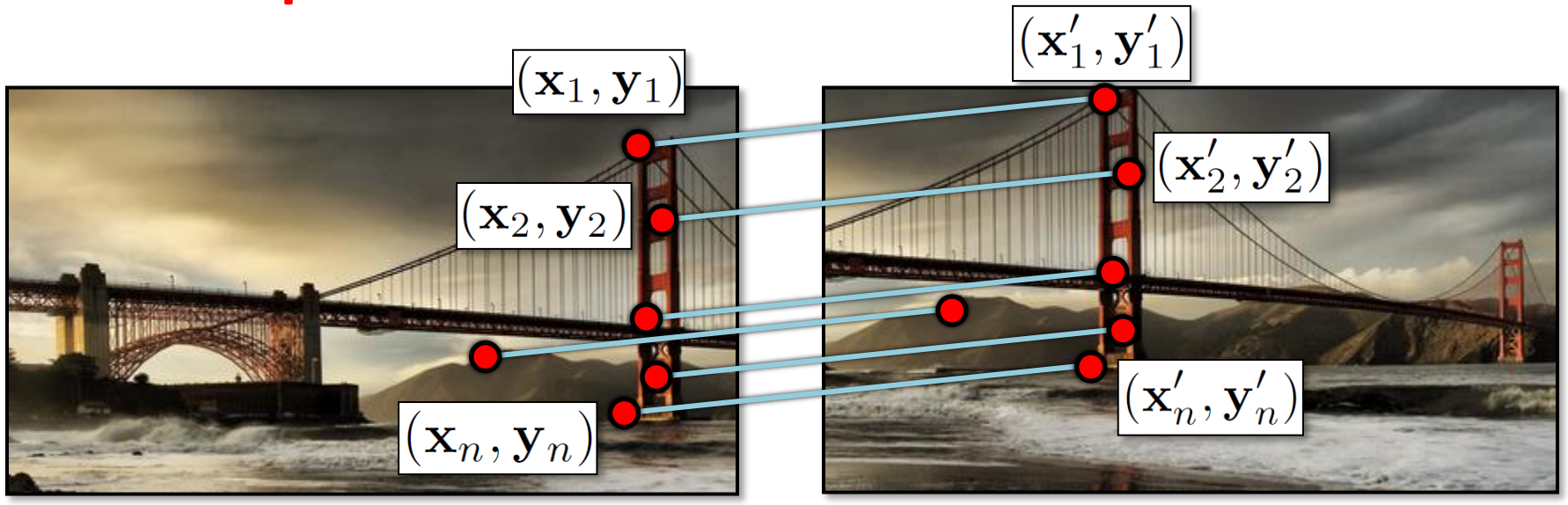




$(\mathbf{x}_t, \mathbf{y}_t)$

How do we solve for ? $(\mathbf{x}_t, \mathbf{y}_t)$

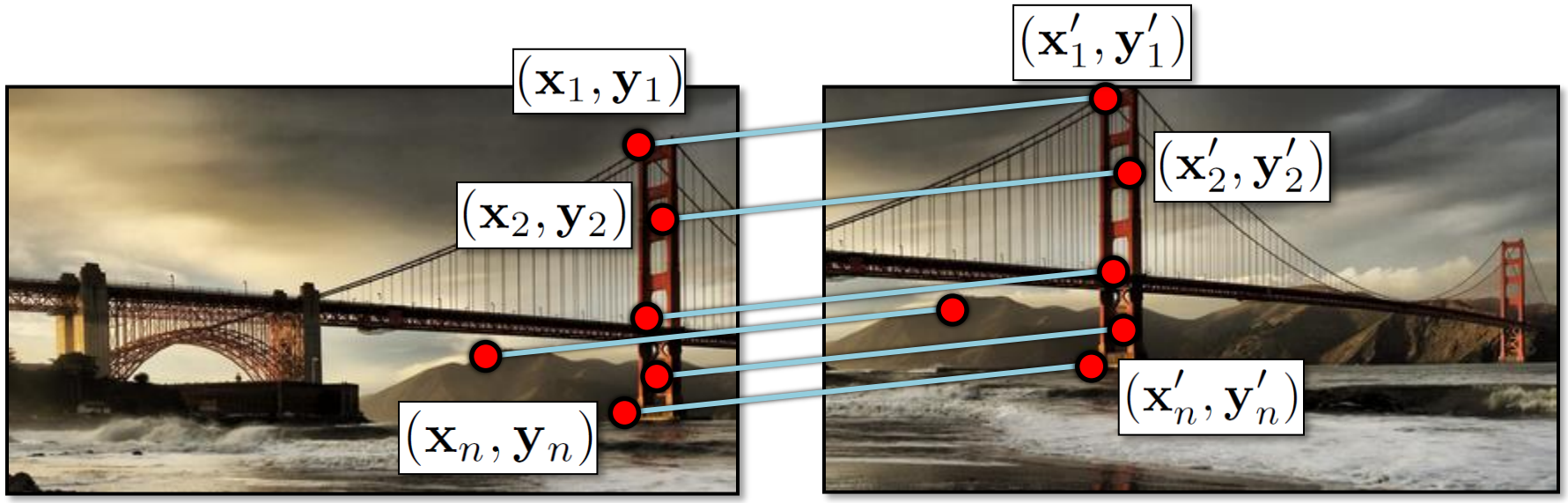
Simple case: translations



Displacement of match *i* = $(\mathbf{x}'_i - \mathbf{x}_i, \mathbf{y}'_i - \mathbf{y}_i)$

$$(\mathbf{x}_t, \mathbf{y}_t) = \left(\frac{1}{n} \sum_{i=1}^{n} \mathbf{x}'_i - \mathbf{x}_i, \frac{1}{n} \sum_{i=1}^{n} \mathbf{y}'_i - \mathbf{y}_i \right)$$

Another view

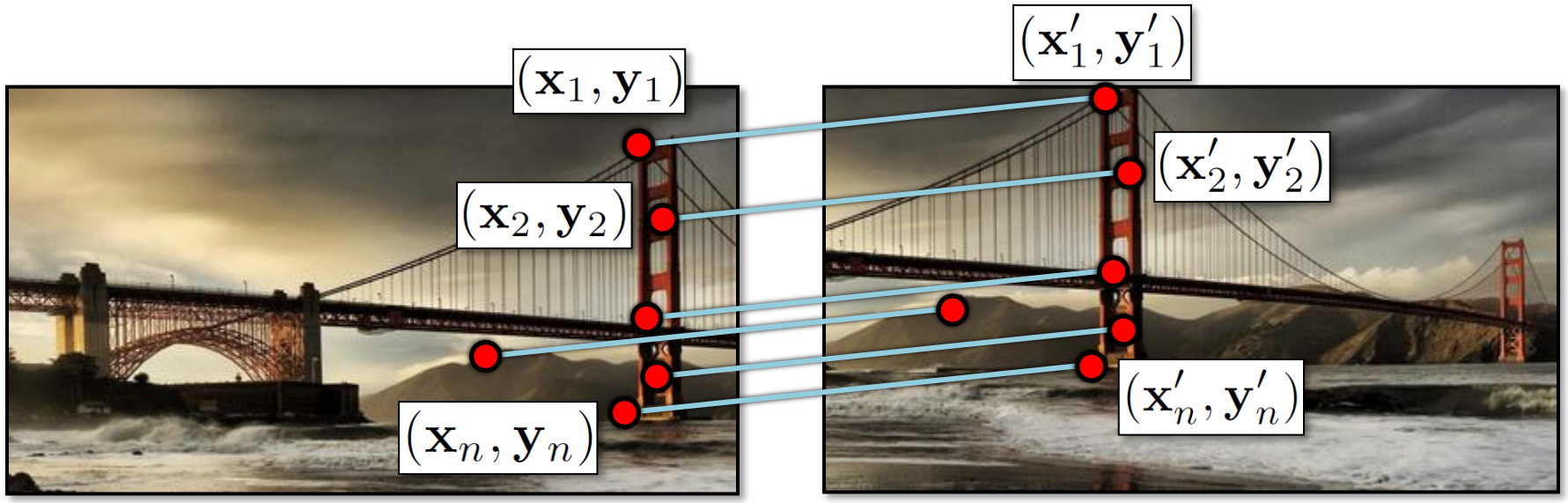


$$\mathbf{x}_i + \mathbf{x_t} = \mathbf{x}'_i$$

$$\mathbf{y}_i + \mathbf{y_t} = \mathbf{y}'_i$$

- System of linear equations
  - What are the knowns? Unknowns?
  - How many unknowns? How many equations (per match)?

Another view



$$\mathbf{x}_i + \mathbf{x_t} = \mathbf{x}'_i$$

$$\mathbf{y}_i + \mathbf{y_t} = \mathbf{y}'_i$$

- Problem: more equations than unknowns
  - "Overdetermined" system of equations
  - We will find the *least squares* solution

Recall Homography Estimation

- $\vec{x}'_{3\times1} = H_{3\times3}\vec{x}_{3\times1}$
- $\begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

- If you have just translation, then, $H = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$, 2 Dof
- So technically, we need one point correspondence to solve for the unknowns as it gives us two equation. But, practically speaking is this robust?

Solution for a System of Linear Equations $A_{M\times N}\vec{x}_{N\times 1} = \vec{b}_{M\times 1}$
Quick Introduction

- The Nonsingular or Determined Case ($M = N$)
- The Underdetermined Case ($M < N$)
- The Overdetermined Case ($M > N$):
  - Least-squares Error

Solution for a System of Linear Equations
Quick Introduction

- Let's suppose we have the following system of linear equation: $A_{M\times N}\vec{x}_{N\times 1} = \vec{b}_{M\times 1}$
- 3 cases to discuss:
  - If the rank of *A* is less than *N* (or $M < N$), we have infinitely many solutions, but we do not want any of them. (This is the underdetermined case.)
  - If rank(*A*) = *N* (also *M* = *N*), we have a unique solution for $\vec{x}$ (determined case): $\vec{x} = A^{-1}\vec{b}$
    - Despite its uniqueness for the solution, we might NOT be interested in this case because of possible noise in the values in *A* and $\vec{b}$.
  - If rank(*A*) = *N* (also *M* > *N*), there does NOT exist a solution for $\vec{x}$.
    - **But this is the case we are interested in, as explained in the next slide**.
    - The best solution will minimize: $\left\|\vec{b} - A\vec{x}\right\|$.
    - The linear least squares solution for $\vec{x}$ must satisfy $A^T\vec{b} = A^T A\vec{x}$ or $A^T A\vec{x} = A^T\ \vec{b}$, which yields $\vec{x} = (A^T A)^{-1}A^T\ \vec{b} = A^{\dagger}\vec{b}$, where $A^{\dagger} = (A^T A)^{-1}A^T$ which is known as the pseudo-inverse of *A*.

Solution for a System of Linear Equations
Quick Introduction: Self Read

- Considering the case ***M > N*** and ***rank(A) = N*** for solving $A\vec{x} = \vec{b}$
  - *A* as an *M*×*N* matrix and $\vec{b}$ as an M-vector. The *N* elements of $\vec{x}$ are the unknowns.
- For this case, obviously, $\vec{b} \in \Re^M$. But, what space does $A\vec{x}$ belong to?
  - Nominally, $A\vec{x}$ also resides in $\Re^M$, since $A\vec{x}$ is an *M*-dimensional vector.
  - More precisely speaking, though, $A\vec{x} \in \Re^N \subset \Re^M$.
    - The reason for that is that the space spanned by the *M* row vectors of *A* is only *N*-dimensional.
    - So the mapping $A\vec{x}$ gives us nominally *M*-dimensional vector but they reside in an N-dimensional subspace of $\Re^M$.
    - It is interesting to reflect on the fact that the dimensionality of the column space is also *N* since we have only *N* column vectors.
    - What this tells us is that the rank of a matrix determines the intrinsic dimensionality of both the row space and the column space.

Solution for a System of Linear Equations

Quick Introduction: Self Read

- We are now faced with the following question: Given that $A\vec{x}$ spans some $\Re^N$ in the $\Re^M$ space in which $\vec{b}$ resides, What is the best solution for $\vec{x}$?
- The best solution for $\vec{x}$ will bring $A\vec{x}$ as close as possible to $\vec{b}$. In other words, the best solution will minimize: $\left\|\vec{b} - A\vec{x}\right\|$.
- Such a solution (called linear least-squares minimization LLSM) would correspond to dropping a perpendicular from $\vec{b} \in \Re^M$ to the subspace spanned by $A\vec{x}$
- Algebraically, the point $\vec{x}$ where the perpendicular from $\vec{b}$ meets the subspace
- spanned by $A\vec{x}$ is given by $A^T\left(\vec{b} - A\vec{x}\right) = \vec{0}$.
- In other words, the linear least squares solution for $\vec{x}$ must satisfy $A^T\vec{b} = A^TA\vec{x}$ or $A^TA\vec{x} = A^T\ \vec{b}$, which yields $\vec{x} = (A^TA)^{-1}A^T\ \vec{b} = A^{\dagger}\vec{b}$, where $A^{\dagger} = (A^TA)^{-1}A^T$ which is known as the pseudo-inverse of *A*.
- Note that: $A^{\dagger}A = (A^TA)^{-1}A^T\ A = I$ (identity Matrix).

Least squares formulation

- For each point $(\mathbf{x}_i, \mathbf{y}_i)$

$$\begin{aligned} \mathbf{x}_i + \mathbf{x_t} &= \mathbf{x}'_i \\ \mathbf{y}_i + \mathbf{y_t} &= \mathbf{y}'_i \end{aligned}$$

- we define the *residuals* as

$$\begin{aligned} r_{\mathbf{x}_i}(\mathbf{x}_t) &= (\mathbf{x}_i + \mathbf{x}_t) - \mathbf{x}'_i \\ r_{\mathbf{y}_i}(\mathbf{y}_t) &= (\mathbf{y}_i + \mathbf{y}_t) - \mathbf{y}'_i \end{aligned}$$

Least squares formulation

- Goal: minimize sum of squared residuals

$$C(\mathbf{x}_t, \mathbf{y}_t) = \sum_{i=1}^{n} \left(r_{\mathbf{x}_i}(\mathbf{x}_t)^2 + r_{\mathbf{y}_i}(\mathbf{y}_t)^2 \right)$$

- “Least squares” solution
- For translations, is equal to mean (average) displacement

Least squares formulation

- Can also write as a matrix equation

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ \vdots & \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_t \\ y_t \end{bmatrix} = \begin{bmatrix} x_1' - x_1 \\ y_1' - y_1 \\ x_2' - x_2 \\ y_2' - y_2 \\ \vdots \\ x_n' - x_n \\ y_n' - y_n \end{bmatrix}$$

$$\underset{2n \times 2}{\mathbf{A}} \quad \underset{2 \times 1}{\mathbf{t}} = \underset{2n \times 1}{\mathbf{b}}$$

Least squares

$$\mathbf{At} = \mathbf{b}$$

- Find **t** that minimizes

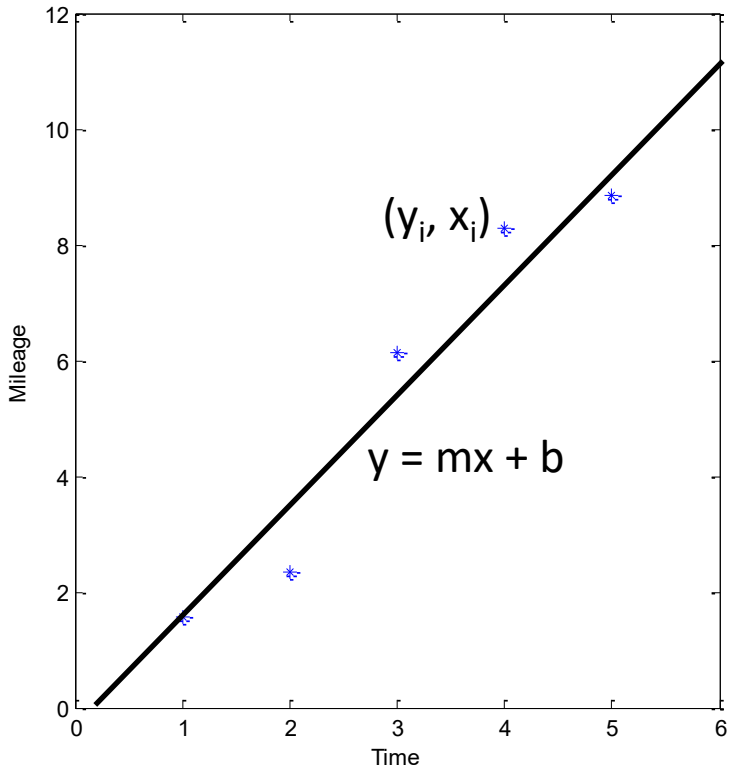
$$||\mathbf{At} - \mathbf{b}||^2$$

- To solve, form the *normal equations*

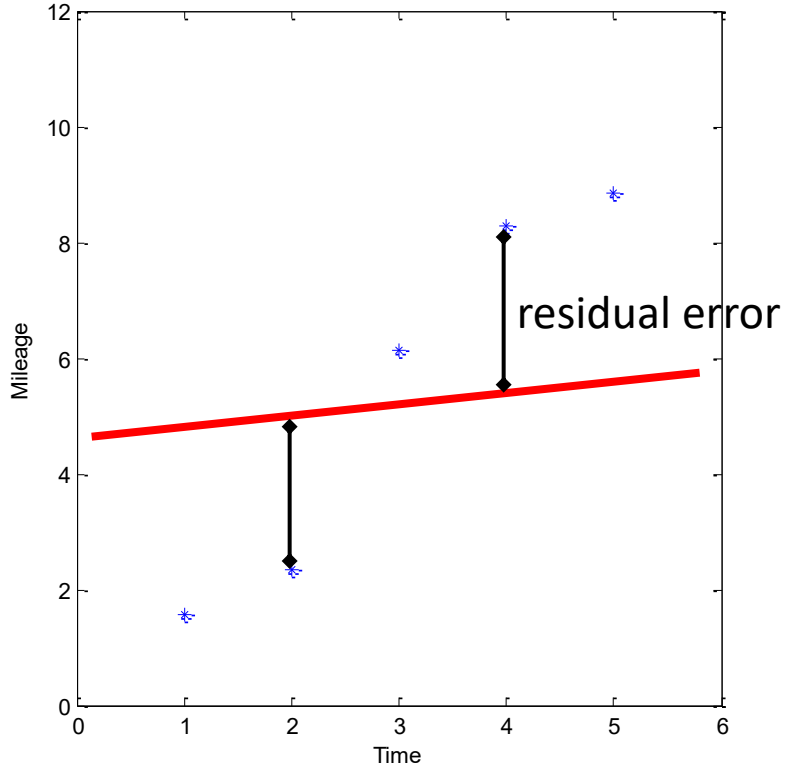
$$\mathbf{A}^{\mathrm{T}}\mathbf{At} = \mathbf{A}^{\mathrm{T}}\mathbf{b}$$

$$\mathbf{t} = \left(\mathbf{A}^{\mathrm{T}}\mathbf{A}\right)^{-1}\mathbf{A}^{\mathrm{T}}\mathbf{b}$$

Aside: Least squares - linear regression



Linear regression



$$\text{Cost}(m, b) = \sum_{i=1}^{n} |y_i - (mx_i + b)|^2$$

Linear regression

$$\begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ \vdots & \\ x_n & 1 \end{bmatrix} \begin{bmatrix} m \\ b \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

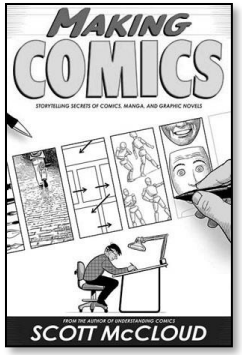
Affine Transformation Case

Rotation + Translation +

Deformation or differential effects

Affine transformations

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$





- How many unknowns or DoF?
- How many equations per match?
- How many matches do we need?

Recall Homography Estimation

- $\vec{x}'_{3\times1} = H_{3\times3}\vec{x}_{3\times1}$
- $\begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

- If you have just Affine Transformation, then, $H = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$, 6 Dof
- So technically, we need three point correspondence to solve for the unknowns as it gives us 6 equation. But, practically speaking is this robust?

Affine transformations

- Residuals:

$$\begin{aligned} r_{x_i}(a,b,c,d,e,f) &= (ax_i + by_i + c) - x'_i \\ r_{y_i}(a,b,c,d,e,f) &= (dx_i + ey_i + f) - y'_i \end{aligned}$$

- Cost function:

$$C(a,b,c,d,e,f) = \sum_{i=1}^{n} \left(r_{x_i}(a,b,c,d,e,f)^2 + r_{y_i}(a,b,c,d,e,f)^2 \right)$$

Affine transformations

- Matrix form

$$\begin{bmatrix} x_1 & y_1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x_1 & y_1 & 1 \\ x_2 & y_2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x_2 & y_2 & 1 \\ & & \vdots & & & \\ x_n & y_n & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x_n & y_n & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \end{bmatrix} = \begin{bmatrix} x'_1 \\ y'_1 \\ x'_2 \\ y'_2 \\ \vdots \\ x'_n \\ y'_n \end{bmatrix}$$

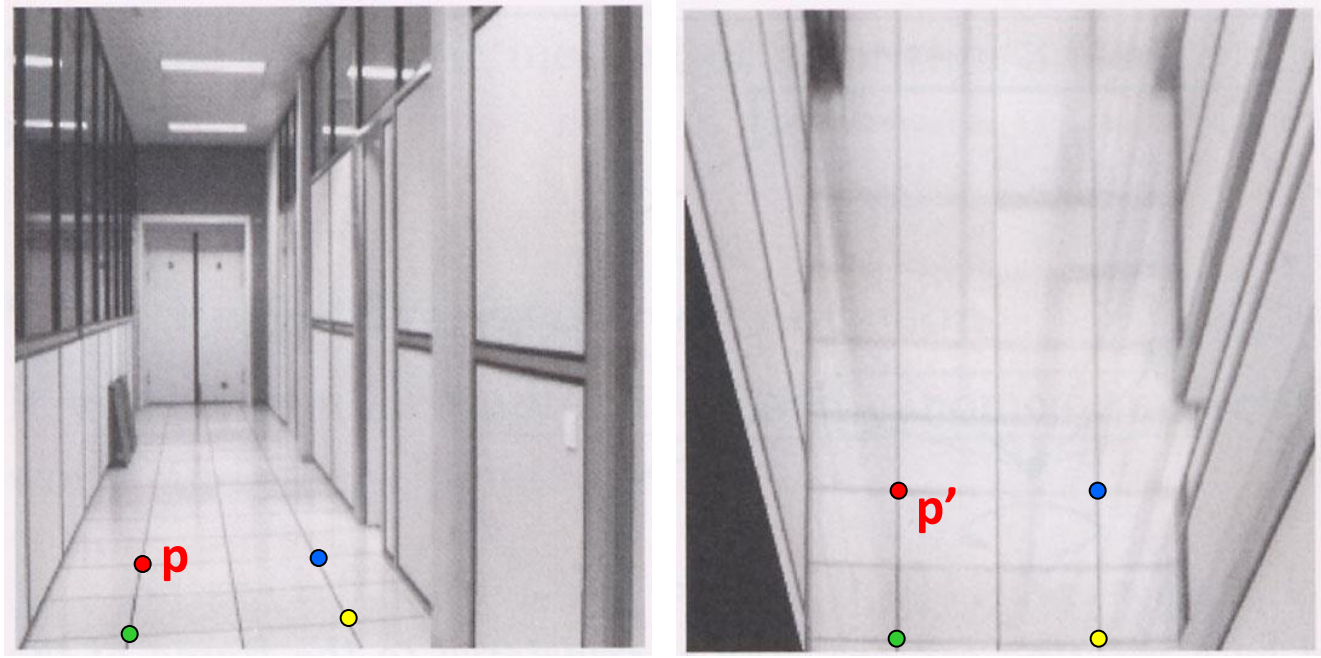
$$\underset{2n \times 6}{\mathbf{A}} \qquad \underset{6 \times 1}{\vec{\boldsymbol{h}}} = \underset{2n \times 1}{\mathbf{b}}$$

Projective Transformation Case

The worst distortion effects

Only straight lines stay straight

Homographies in General: Projective Transformation



To unwarp (rectify) an image

- solve for homography **H** given **p** and **p'**
- solve equations of the form: **p'** = **Hp**
  - linear in unknowns: coefficients of **H**
  - H is defined up to an arbitrary scale factor
  - how many matches are necessary to solve for **H**?

Solving for homographies

$$\begin{bmatrix} x_i' \\ y_i' \\ 1 \end{bmatrix} \cong \begin{bmatrix} h_{00} & h_{01} & h_{02} \\ h_{10} & h_{11} & h_{12} \\ h_{20} & h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}$$

$$x_i' = \frac{h_{00}x_i + h_{01}y_i + h_{02}}{h_{20}x_i + h_{21}y_i + h_{22}}$$

$$y_i' = \frac{h_{10}x_i + h_{11}y_i + h_{12}}{h_{20}x_i + h_{21}y_i + h_{22}}$$

Not linear!

$$x_i'(h_{20}x_i + h_{21}y_i + h_{22}) = h_{00}x_i + h_{01}y_i + h_{02}$$
$$y_i'(h_{20}x_i + h_{21}y_i + h_{22}) = h_{10}x_i + h_{11}y_i + h_{12}$$

Solving for homographies

$$
\begin{aligned}
x_i'(h_{20}x_i + h_{21}y_i + h_{22}) &= h_{00}x_i + h_{01}y_i + h_{02} \\
y_i'(h_{20}x_i + h_{21}y_i + h_{22}) &= h_{10}x_i + h_{11}y_i + h_{12}
\end{aligned}
$$

$$
\begin{bmatrix}
x_i & y_i & 1 & 0 & 0 & 0 & -x_i'x_i & -x_i'y_i & -x_i' \\
0 & 0 & 0 & x_i & y_i & 1 & -y_i'x_i & -y_i'y_i & -y_i'
\end{bmatrix}
\begin{bmatrix}
h_{00} \\ h_{01} \\ h_{02} \\ h_{10} \\ h_{11} \\ h_{12} \\ h_{20} \\ h_{21} \\ h_{22}
\end{bmatrix}
=
\begin{bmatrix}
0 \\ 0
\end{bmatrix}
$$

Solving for homographies

$$\begin{bmatrix} x_1 & y_1 & 1 & 0 & 0 & 0 & -x'_1x_1 & -x'_1y_1 & -x'_1 \\ 0 & 0 & 0 & x_1 & y_1 & 1 & -y'_1x_1 & -y'_1y_1 & -y'_1 \\ & & & & & \vdots & & & \\ x_n & y_n & 1 & 0 & 0 & 0 & -x'_nx_n & -x'_ny_n & -x'_n \\ 0 & 0 & 0 & x_n & y_n & 1 & -y'_nx_n & -y'_ny_n & -y'_n \end{bmatrix} \begin{bmatrix} h_{00} \\ h_{01} \\ h_{02} \\ h_{10} \\ h_{11} \\ h_{12} \\ h_{20} \\ h_{21} \\ h_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}$$

$\mathbf{A}$ (2n × 9) $\quad \mathbf{h}$ (9) $\quad \mathbf{0}$ (2n)

Defines a least squares problem: minimize $\|\mathbf{A}\mathbf{h} - \mathbf{0}\|^2$

- Since $\mathbf{h}$ is only defined up to scale, solve for unit vector $\hat{\mathbf{h}}$
- Solution: $\hat{\mathbf{h}}$ = eigenvector of $\mathbf{A}^T\mathbf{A}$ with smallest eigenvalue
- Works with 4 or more points

Recap: Two Common Optimization Problems

Problem statement

$$\text{minimize} \quad \|\mathbf{Ax} - \mathbf{b}\|^2$$

(least squares solution to $\mathbf{Ax} = \mathbf{b}$)

Solution

$$\mathbf{x} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{b}$$

$\mathbf{x} = \mathbf{A}\backslash\mathbf{b}$ (Left Division)

Problem statement

$$\text{minimize} \quad \mathbf{x}^T\mathbf{A}^T\mathbf{Ax} \quad \text{s.t. } \mathbf{x}^T\mathbf{x} = 1$$

(non−trivial lsq solution to $\mathbf{Ax} = 0$)

Solution (i.e. Null Space)

$$[\mathbf{v}, \lambda] = \text{eig}(\mathbf{A}^T\mathbf{A})$$

$\lambda_1 < \lambda_{2..n}$: $\mathbf{x} = \mathbf{v}_1$

Computing transformations

A



B







Image Alignment Problem

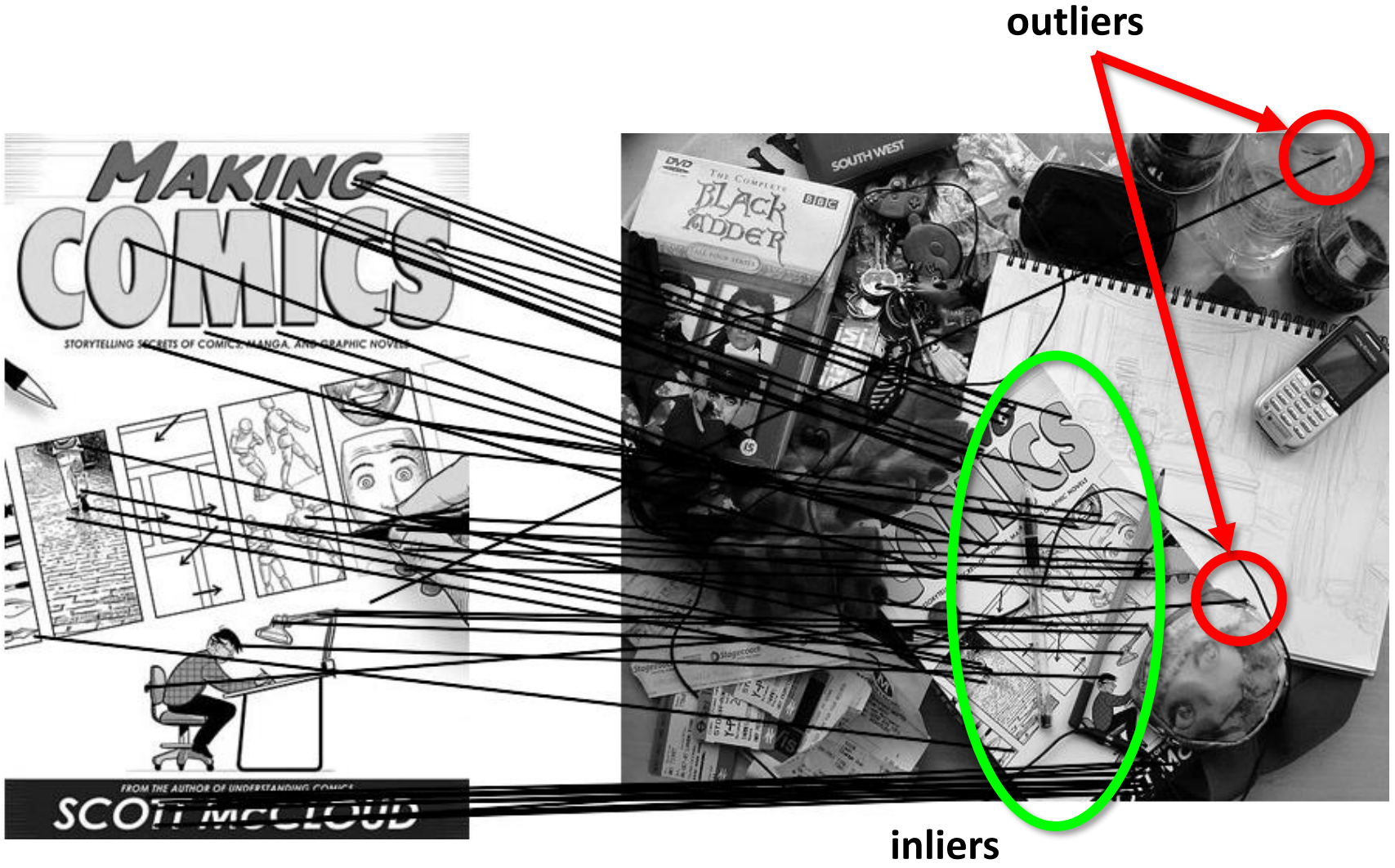
Image alignment algorithm

Given images A and B

1. Compute image features for A and B
2. Match features between A and B
3. Compute homography between A and B using least squares on set of matches

What could go wrong?

Outliers


outliers
MAKING
COMICS
STORYTELLING SECRETS OF COMICS, MANGA, AND GRAPHIC NOVELS
FROM THE AUTHOR OF UNDERSTANDING COMICS
SCOTT McCLOUD
DVD
THE COMPLETE
BLACK ADDER
BBC
SOUTH WEST
inliers

Questions?